

REGION OF WATERLOO INTL, ON, Canada

WMO: 713680

Lat: 43.4616N

Lon: 80.3856W

Elev: 322

StdP: 97.52

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.9	(c) -17.6	(d) -24.5	(e) 0.4	(f) -20.0	(g) -21.3	(h) 0.6	(i) -17.2	(j) 13.1	(k) -5.7	(l) 12.0	(m) -6.4	(n) 2.5	(o) 310	(p) 0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	30.5	21.8	28.9	21.1	27.4	20.4	23.3	28.3	22.5	27.0	21.6	25.6	5.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.2	25.8	21.0	16.3	24.8	20.1	15.4	23.9	71.3	28.2	67.7	27.1	64.4	25.6	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	10.0	8.9	-25.4	32.8	4.4	1.7	-28.6	34.0	-31.2	35.0	-33.6	36.0	-36.8	37.2
			-25.6	25.2	4.4	1.2	-28.8	26.0	-31.3	26.7	-33.8	27.4	-37.0	28.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-6.5	-6.1	-0.9	6.1	13.0	17.8	20.4	19.2	15.7	9.0	2.8	-2.6
	DBStd	10.62	6.47	6.02	6.19	4.85	4.65	3.64	3.07	2.96	3.86	4.55	4.66	4.92
	HDD10.0	2146	510	451	341	136	21	0	0	0	2	76	219	391
	HDD18.3	4189	768	684	596	368	178	52	14	25	97	292	466	649
	CDD10.0	1197	0	0	4	18	113	234	321	287	174	44	3	0
	CDD18.3	199	0	0	0	0	12	37	77	53	19	1	0	0
	CDH23.3	1941	0	0	2	7	142	376	751	456	192	16	0	0
	CDH26.7	484	0	0	0	0	28	106	210	96	43	2	0	0
(14) Wind	WSAvg	4.0	4.8	4.7	4.5	4.7	3.9	3.5	3.1	3.0	3.1	3.9	4.3	4.7
(15) Precipitation	PrecAvg	850	73	59	54	73	81	78	89	66	73	75	72	59
	PrecMax	1073	120	111	131	130	156	169	172	130	155	137	145	113
	PrecMin	576	33	28	11	23	20	27	15	7	21	17	16	14
	PrecStd	127	25	23	30	30	32	36	47	30	37	33	30	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	11.7	20.2	24.3	29.4	31.6	32.9	31.6	30.4	25.4	17.5	11.2
		MCWB	10.5	9.7	15.4	15.8	20.1	22.2	22.8	22.4	22.6	18.8	13.2	10.1
	2%	DB	7.4	7.1	14.2	20.8	27.1	29.7	30.8	29.0	27.8	22.0	14.5	8.2
		MCWB	6.4	5.5	10.4	13.6	19.4	21.2	21.9	21.4	20.7	16.8	11.5	7.1
	5%	DB	4.3	3.7	11.0	17.9	24.6	27.5	29.0	27.5	25.6	19.2	12.3	5.7
		MCWB	3.3	2.1	8.2	11.5	17.9	20.3	21.2	20.9	19.5	15.4	9.9	4.7
	10%	DB	2.1	1.7	7.9	15.0	22.0	25.2	27.3	26.0	23.3	16.7	10.1	3.6
		MCWB	1.2	0.4	5.4	9.8	15.9	18.9	20.2	20.0	18.2	13.9	8.2	2.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	10.6	15.8	16.5	22.0	23.6	24.7	24.6	23.6	20.4	14.3	10.5
		MCDB	10.7	11.4	19.9	21.3	27.4	29.2	30.1	27.6	28.5	23.7	15.9	11.2
	2%	WB	6.2	5.7	11.2	14.8	20.3	22.4	23.5	22.9	22.0	17.8	12.0	7.2
		MCDB	6.8	7.2	13.1	19.4	25.7	27.9	28.4	27.3	26.2	20.2	13.6	7.9
	5%	WB	3.2	2.2	8.4	12.9	18.7	21.2	22.5	21.9	20.6	15.9	10.0	4.8
		MCDB	4.1	3.5	10.7	16.6	23.1	25.7	27.0	25.5	23.8	18.9	11.9	5.7
	10%	WB	1.2	0.5	5.5	10.4	17.0	20.0	21.5	21.0	19.2	14.2	8.3	2.7
		MCDB	1.9	1.6	7.6	14.2	20.8	24.0	25.6	24.5	22.3	16.2	10.1	3.4
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.4	9.1	11.0	11.9	11.7	12.1	11.8	12.0	9.8	8.0	6.2
		MCDBR	8.3	10.3	14.0	16.5	15.0	14.3	14.0	13.5	14.8	14.1	12.1	8.3
	5% WB	MCWBR	8.1	8.8	10.1	10.4	8.4	7.5	6.7	6.8	8.1	9.1	9.3	7.7
		MCWBR	8.2	9.5	13.1	14.3	13.4	12.2	11.9	10.8	12.1	12.6	10.7	8.1
(40) Clear-Sky Solar Irradiance	taub		0.286	0.332	0.326	0.363	0.387	0.419	0.411	0.399	0.368	0.351	0.305	0.286
		taud	2.371	2.188	2.391	2.333	2.309	2.235	2.277	2.298	2.377	2.412	2.505	2.438
	Ebn at Noon		846	848	915	902	886	854	857	855	857	825	824	816
		Edh at Noon	78	112	104	120	127	137	130	123	105	90	69	67
(44) All-Sky Solar Radiation	RadAvg		1.41	2.23	3.50	4.44	5.38	5.94	5.91	5.16	4.14	2.48	1.53	1.02
	RadStd		0.18	0.28	0.36	0.51	0.52	0.40	0.40	0.26	0.40	0.23	0.21	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page